

3. Magnetic vector potential

In the previous section, we found that $\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r})$, where $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{r} dV'$. We have shown that $\nabla \cdot \vec{A}(\vec{r}) = 0$. To satisfy $\nabla \times \vec{A}(\vec{r}) = \vec{B}(\vec{r})$, $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{r} dV'$ is only one possible, divergence-less ($\nabla \cdot \vec{A}(\vec{r}) = 0$) choice. There exists an infinite number of $\vec{A}$ to satisfy $\nabla \times \vec{A}(\vec{r}) = \vec{B}(\vec{r})$. To see this, set $\vec{A}' = \vec{A} + \nabla\phi$ where ϕ is an arbitrary scalar function,

$$\nabla \times \vec{A}' = \nabla \times \vec{A} + \nabla \times (\nabla\phi) = \nabla \times \vec{A} = \vec{B}.$$

The vector potential $\vec{A}$, like the scalar potential in electrostatics, is not uniquely determined. We will explore its fundamental significance in greater detail later in this chapter.

What is the physical meaning of $\vec{A}$? Starting from $\vec{B} = \nabla \times \vec{A}$, and integrate $\vec{B}$ on an open surface, we get:

$$\underbrace{\int_S \vec{B} \cdot d\vec{a}}_{\Phi_B: \text{Magnetic flux}} = \int_S \nabla \times \vec{A} \cdot d\vec{a} = \oint_P \vec{A} \cdot d\vec{l}.$$

The line integral of $\vec{A}$ along a closed loop is the total flux of $\vec{B}$ enclosed by the loop, piecing through the open surface attached to the loop. See the following figure.

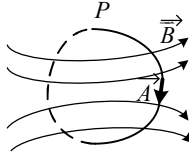


Figure. Relation between $\vec{B}$ and $\vec{A}$.

Note, if we compare the divergence and curl of $\vec{A}$,

$$\begin{cases} \nabla \times \vec{A} = \vec{B} \\ \nabla \cdot \vec{A} = 0 \end{cases},$$

to those of $\vec{B}$,

$$\begin{cases} \nabla \times \vec{B} = \mu_0 \vec{J} \\ \nabla \cdot \vec{B} = 0 \end{cases},$$

we observe that the relationship between $\vec{A}$ and $\vec{B}$ mirrors that between $\vec{B}$ and $\vec{J}$. For example, the magnetic field generated by a straight current-carrying wire forms concentric circles around the wire, just as the vector potential of a magnetic field confined within a straight wire (such as in a long solenoid) forms concentric loops around the solenoid. See the first row of the following figure. The magnetic field lines generated by a current loop and the vector potential lines of a magnetic field loop (confined within a toroidal coil) also show similarity. See the second row of the figure.

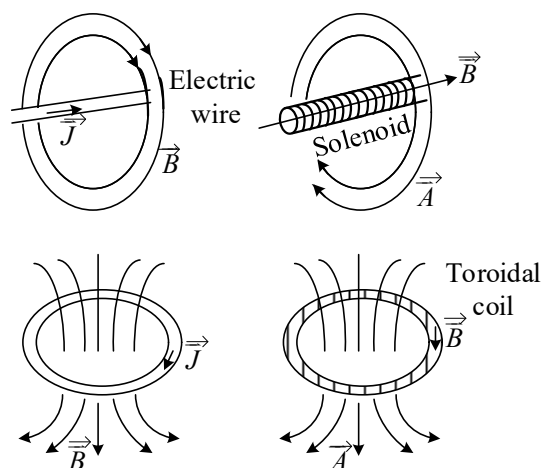


Figure. Analogous relationships: $\vec{B}$ encircles $\vec{J}$, just as $\vec{A}$ encircles $\vec{B}$.

Top: straight wires; Bottom: circular loops.

Although the introduction of the vector potential $\vec{A}$ may seem redundant—since, unlike the scalar potential, it does not always simplify calculations compared to using the magnetic field $\vec{B}$ directly—it plays a fundamental role in physics beyond being just a mathematical convenience. In classical electromagnetism, $\vec{A}$ provides a framework for expressing Maxwell's equations in a more unified form, particularly in the Lorenz gauge (see Chapter 5), facilitating solutions in electrodynamics. More importantly, in quantum mechanics, the vector potential acquires direct physical significance, as demonstrated by the Aharonov-Bohm (A-B) effect, where charged particles exhibit phase shifts due to $\vec{A}$, even in regions where $\vec{B}$ is zero (we will discuss this effect in more detail later in this Chapter). This experiment confirms that $\vec{A}$ is not merely an auxiliary quantity but an essential entity that influences quantum wavefunctions, reinforcing the fundamental role of gauge fields in modern physics, including quantum electrodynamics and gauge theories.

4. Magnetic fields in matter

When thinking about magnetism, most people first picture a permanent magnet. However, everything we have learned so far seems unrelated to it. We've established that magnetic fields originate from steady currents, yet a magnet appears to have no external power source. So, how does a current come into play? Here, we will answer this question. We will explore how materials respond to externally applied magnetic fields and how certain materials can generate magnetic fields. Just as we previously examined the interaction between the electric field and an insulator by introducing the concept of an electric dipole, we will now begin with the idea of a magnetic dipole.

1) Magnetic dipole

Magnetism in matter arises from tiny current loops, primarily due to electron spin and orbital motion. Such current loops, when viewed from a distance, can be treated as ideal magnetic dipoles. To analyze the magnetic potential generated by a current loop, we refer to the following figure, where a current loop (in the general case, it does not have to be perfectly circular) is centered at the origin.

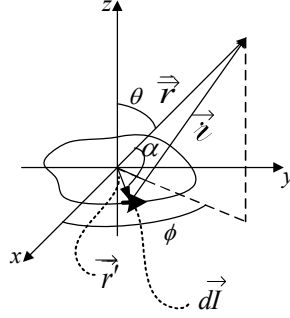


Figure. A current loop centered at the origin.

We can compute the vector potential $\vec{A}$ of this magnetic loop by integrating the contribution from the current element $d\vec{l}$, where the position of $d\vec{l}$ is denoted by $\vec{r}'$, the field point is $\vec{r}$, and the separation vector is $\vec{z}$. The angle between $\vec{r}$ and $\vec{r}'$ is denoted as α . According to the Law of Cosines, the length of $\vec{z}$ is $\sqrt{r^2 + r'^2 - 2rr' \cos \alpha} = r \sqrt{1 - 2 \left(\frac{r'}{r}\right) \cos \alpha + \left(\frac{r'}{r}\right)^2}$, so we can apply the Generating Function of the Legendre Polynomial:

$$\frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad (t < 1)$$

We can readily get

$$\frac{1}{z} = \frac{1}{r} \sum_{n=0}^{\infty} \left(\frac{r'}{r}\right)^n P_n(\cos \alpha).$$

Now we can perform the integral of the vector potential

$$\begin{aligned} \vec{A}(\vec{r}) &= \frac{\mu_0 I}{4\pi} \oint \frac{d\vec{l}'}{z} = \frac{\mu_0 I}{4\pi} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \oint (r')^n P_n(\cos \alpha) d\vec{l}' \\ &= \frac{\mu_0 I}{4\pi} \left[\underbrace{\frac{1}{r} \oint d\vec{l}'}_{\substack{=0 \\ \text{(no monopole)}}} + \frac{1}{r^2} \oint r' \cos \alpha d\vec{l}' + \underbrace{\dots}_{\substack{\text{higher} \\ \text{order} \approx 0}} \right] \\ &\approx \frac{\mu_0 I}{4\pi r^2} \oint (\hat{r} \cdot \vec{r}') d\vec{l}' \end{aligned}$$

The above formula may not appear familiar, so we need to transform it into a form similar to $\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$. To achieve this, we simplify the situation by: (i) Assuming the loop is a circle. (ii) Rotating the coordinate system to align $\vec{r}$ in the $y-z$ plane. Refer to the following figure, which illustrates $d\vec{l}'$ in the $x-y$ plane.

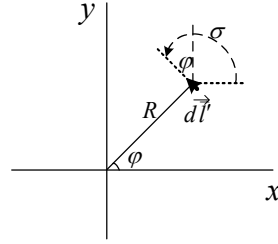


Figure. $d\vec{l}'$ displayed in the $x-y$ plane.

Taking advantage of the fact that $r' = R$ is a fixed length and $d\vec{l}'$ is a two-dimensional vector in the $x-y$ plane, we use a complex number to represent $d\vec{l}'$ to simplify our derivation:

$$d\vec{l}' = \underbrace{(R d\varphi)}_{\text{modulus}} \cdot e^{j\sigma}$$

where σ is the angle of the vector. According to the above figure, $\sigma = \varphi + \pi/2$, leading to

$d\vec{l}' = (R d\varphi) \cdot e^{j(\varphi+\pi/2)} = j(R d\varphi) \cdot e^{j\varphi}$. To assist our analysis, we re-plot the circular current loop in Cartesian coordinates in the following figure.

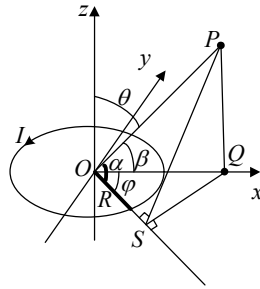


Figure. A circular current loop centered at the origin.

Geometric relationship shown in the figure helps us to demonstrate $\cos \alpha = \cos \beta \cdot \cos \varphi$ by observing that $\frac{OS}{OP} = \frac{OS}{OQ} \cdot \frac{OQ}{OP}$. Remember the key to calculating the vector potential, according to our analysis above, is to evaluate $\oint r' \cos \alpha d\vec{l}'$, by substituting $\cos \alpha$ and $d\vec{l}'$ using the expressions we just obtained,

$$\begin{aligned}
\oint r' \cos \alpha d\vec{l}' &= jR^2 \cos \beta \int_0^{2\pi} \cos \varphi e^{j\varphi} d\varphi \\
&= jR^2 \sin \theta \int_0^{2\pi} \cos \varphi (\cos \varphi + j \sin \varphi) d\varphi \\
&= jR^2 \sin \theta \cdot \frac{1}{2} \cdot 2\pi = j(\pi R^2) \sin \theta
\end{aligned}$$

In the formula above, the imaginary unit j represents a unit vector pointing along the y -direction. This, along with $\sin \theta$, is reminiscent of a cross product between $\hat{z}$ and $\hat{r}$. Additionally, πR^2 represents the area of the loop. In summary, we can concisely re-write the result as

$$A(\vec{r}) = \frac{\mu_0 I \vec{a} \times \hat{r}}{4\pi r^2},$$

where $\vec{a}$ is a vector perpendicular to the x - y plane, which follows the right-hand rule with the current. Its magnitude, $|\vec{a}| = \pi R^2$, represents the area of the loop. In the above formula, define the magnetic dipole moment as $I\vec{a} = \vec{m}$, we reach a formula which is analogous to the potential of an electric dipole ($\phi(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$):

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}.$$

The above formula can be written in the spherical coordinates as

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \cdot \frac{m \sin \theta}{r^2} \hat{\phi}.$$

Based on this formula, we are able to compute the magnetic field due to such a magnetic dipole,

$$\begin{aligned}
\vec{B} = (\nabla \times \vec{A}) &= \frac{\mu_0 m}{4\pi r \sin \theta} \cdot \left[\frac{\partial}{\partial \theta} \left(\frac{\sin^2 \theta}{r^2} \right) \right] \hat{r} - \frac{\mu_0 m}{4\pi} \frac{1}{r} \frac{\partial}{\partial r} \left(r \cdot \frac{\sin \theta}{r^2} \right) \hat{\theta} \\
&= \frac{\mu_0 m}{4\pi r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}),
\end{aligned}$$

which has exactly the same field pattern as an electric dipole (which I have asked you to compute in the homework):

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}).$$

This is an absolutely amazing result!

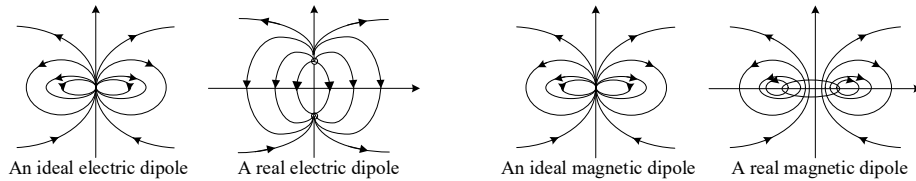


Figure. Field lines of ideal and real electric (E field) and magnetic (B field) dipoles.

2) Material magnetization

When a substance is subject to an external magnetic field, all the electrons in the material will feel a torque. In the following figure, the electron spin is modelled as a small magnetic dipole – for visualization purposes, the spin is represented as a current loop with the current directions labelled using arrows. The torque tends to align the magnetic dipole moment with the external magnetic field.

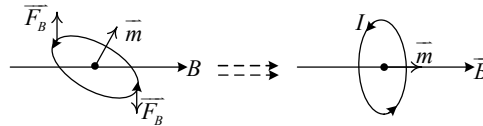


Figure. The dipole moment generated by electron spin experiences a torque and tends to align with the external magnetic field.

However, there is another effect arising from the orbital motion of electrons around nuclei. Refer to the following figure. The arrows along the loop indicate the rotation direction of an electron. In the presence of an external magnetic field oriented as shown, the current loop experiences a torque. However, the orbital motion is much more rigid than the spin direction and is difficult to tilt. Instead, the electron "feels" a net force toward the center of its rotation, accelerating its motion and increasing the dipole moment in a direction opposing the external magnetic field. The magnitude of this effect is much smaller than the first effect due to spin re-orientation.

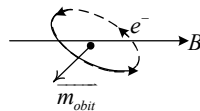


Figure. The dipole moment arising from electron orbital motion tends to develop an additional component opposite to the external magnetic field.

In summary, the process by which a material acquires an overall magnetic dipole moment is known as material magnetization. There are two primary contributions to magnetization:

- Electron spin tends to generate a magnetization parallel to the external magnetic field.
- Electron orbital motion typically induces a magnetization opposing the external magnetic field.

Both effects coexist in any material, and the overall magnetic response is determined by which effect dominates. If all electrons in the material are paired, their net spin dipole moment cancels due to the Pauli exclusion principle, leaving orbital motion as the dominant effect. Such materials are known

as diamagnetic materials, which develop a magnetization opposing the external field. In contrast, if each atom or molecule has an unpaired electron (or multiple unpaired electrons with the same spin—for example, the oxygen molecule has two unpaired electrons with parallel spins), the material tends to develop a magnetization parallel to the external magnetic field. These materials are classified as paramagnetic materials. This distinction between diamagnetism and paramagnetism is fundamental to understanding the magnetic behavior of different substances.

Keep in mind that our explanations of material magnetization above are intuitive but not fully rigorous. The true nature of magnetization is governed by quantum mechanics, which is beyond the scope of this course. In addition to paramagnetism and diamagnetism, other, more complex and nonlinear magnetic responses exist, such as ferromagnetism and ferrimagnetism, which we will briefly discuss in a separate section.

Now we will study magnetization using a more quantitative approach. Recall that we established the framework of material polarization by defining a key quantity, the polarization vector $\vec{P}$, which represents the electric dipole moment density (total dipole moment per unit volume). Once again, the parallelism between electricity and magnetism is evident in our treatment of magnetization. We follow almost the same approach by defining a fundamental parameter, the magnetization vector $\vec{M}$, which presents the magnetic dipole moment density:

$$\vec{M} = \frac{\text{Total magnetic dipole moment in } dV}{dV}.$$

Note that the unit of magnetic dipole moment $\vec{m}$ is $[A] \cdot [m^2]$, while the unit for magnetization $\vec{M}$ is $[A] \cdot \frac{[m^2]}{[m^3]} = \frac{[A]}{[m]}$, the same as surface current density.

3) Bound currents

To aid our study, let's review the concept of bound charge in electrostatics:

$$\rho_b = -\nabla \cdot \vec{P}, \sigma_b = \vec{P} \cdot \hat{n}.$$

In simple terms, a polarized material tends to develop bound charges, which in turn generate their own electric field, influencing the total electric field.

By analogy, it is natural to expect that a magnetized material will develop bound currents (both volume and surface currents), which contribute an additional magnetic field, thereby modifying the total magnetic field.

To derive the formulae for the bound currents, we start from extending the magnetic potential due to a single magnetic dipole,

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2},$$

into one that is generated by all the dipoles within a magnetized material, the potential can be directly computed by a volume integral:

$$\begin{aligned}\vec{A}(\vec{r}) &= \frac{\mu_0}{4\pi} \int \frac{\vec{M}(\vec{r}') \times \hat{z}}{z^2} dV' \\ &= \frac{\mu_0}{4\pi} \int \left[\vec{M}(\vec{r}') \times \left(\nabla' \frac{1}{z} \right) \right] dV' \\ &= \frac{\mu_0}{4\pi} \left[\int \frac{1}{z} [\nabla' \times \vec{M}(\vec{r}')] dV' - \int \nabla' \times \left[\frac{\vec{M}(\vec{r}')}{z} \right] dV' \right].\end{aligned}$$

We realize that $\int \nabla' \times \left[\frac{\vec{M}(\vec{r}')}{z} \right] dV'$ is in the form of $\int (\nabla \times \vec{v}) dV$, which presents the volume integral of a curl. According to the generalized Stokes' theorem, this integral can be converted into a surface integral of the function $\vec{v}$. The identity, analogous to both the Divergence Theorem and Stokes' Theorem, can be expressed in the following form:

$$\int_V (\nabla \times \vec{v}) dV = - \oint_S \vec{v} \times d\vec{a}.$$

This identity leads to our final result

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{1}{z} \underbrace{[\nabla' \times \vec{M}(\vec{r}')] dV'}_{\vec{J}_b} + \frac{\mu_0}{4\pi} \oint_S \frac{1}{z} \underbrace{(\vec{M}(\vec{r}') \times \hat{n}') d\vec{a}'}_{\vec{K}_b}.$$

Recall that for a volume current distribution,

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{z} dV',$$

while for a surface current distribution,

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{K}(\vec{r}')}{z} d\vec{a}',$$

we conclude that the vector potential generated by magnetization consists of two contributions: one from the volume bound current:

$$\boxed{\vec{J}_b = \nabla \times \vec{M}(\vec{r})}$$

and the other from the surface bound current:

$$\boxed{\vec{K}_b = \vec{M}(\vec{r}) \times \hat{n}}.$$

These expressions closely resemble those of bound charges in electrostatics. Note that if a cylindrical material is magnetized along its long axis, the surface bound current will flow in the azimuthal direction, forming closed loops around the cylinder. Its direction follows the right-hand

rule with respect to the magnetization direction, as shown in the following figure.

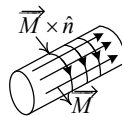


Figure. Directions of magnetization and the bound current.

4) The $\vec{H}$ field

Suppose a piece of material is placed inside a current-carrying solenoid. As we will later learn, such a solenoid produces an almost uniform magnetic field along its axis. The material will become magnetized either in the direction of the field (if it is paramagnetic) or opposite to the field (if it is diamagnetic).

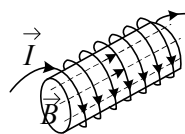


Figure. A method for inducing magnetization in a material.

Now, we have two types of currents: free current, which is externally applied and controlled, and bound current, which is induced within the material and depends on its response:

$$\vec{J} = \vec{J}_f + \vec{J}_b.$$

Substituting the above equation into Ampere's Law, we have:

$$\frac{1}{\mu_0} (\nabla \times \vec{B}) = \vec{J} = \vec{J}_f + \vec{J}_b = \vec{J}_f + (\nabla \times \vec{M}).$$

Merging the two curl terms gives:

$$\nabla \times \left(\frac{1}{\mu_0} \vec{B} - \vec{M} \right) = \vec{J}_f.$$

Here, we introduce an auxiliary vector:

$$\boxed{\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}},$$

which allows us to express the equation in a form resembling Ampère's Law, but involving only the free current:

$$\boxed{\nabla \times \vec{H} = \vec{J}_f}.$$

Note, the naming of $\vec{B}$ and $\vec{H}$ varies across different textbooks. Some sources refer to $\vec{B}$ as the “magnetic induction”, or “magnetic flux density”, and $\vec{H}$ as the “magnetic field”, “auxiliary field”, “magnetic field intensity”, or “magnetic field strength”. Following Griffiths' convention, we will

refer to $\vec{B}$ as the magnetic field throughout our discussion. Integrating the above Ampere's Law for $\vec{H}$ over an open surface S with boundary P and applying Stokes' Theorem, we obtain:

$$\oint_P \vec{H} \cdot d\vec{l} = I_{f,enc},$$

where $I_{f,enc}$ represents the total free current enclosed by the loop P .

As we discussed earlier, both paramagnetism and diamagnetism are extremely weak, so in the context of these two types of magnetization, the material's response is linear. Consequently, by convention, we can write

$$\begin{aligned}\vec{B} &= \mu_0(\vec{H} + \vec{M}) \\ &= \mu_0\vec{H} + \mu_0\chi_m\vec{H} \\ &= \mu_0(1 + \chi_m)\vec{H} \\ &= \mu_0\mu_r\vec{H} = \mu\vec{H}\end{aligned}$$

where the unitless quantity χ_m is called the magnetic susceptibility, a small number typically ranging from 10^{-9} to 10^{-3} . It is positive for paramagnetic materials and negative for diamagnetic materials. The relative permeability μ_r is another dimensionless parameter, while μ , known as the permeability, has the SI unit of henry per meter (H/m).

5. Boundary conditions

To derive the magnetic boundary conditions, we use the same schematic approach as in the derivation of electric boundary conditions.

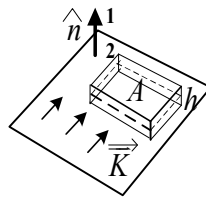


Figure: Schematic illustration for deriving boundary condition for the normal component.

1) Normal (perpendicular) components

The perpendicular component of the magnetic field $\vec{B}$ is always continuous across a boundary. This follows from Gauss's Law for Magnetism, which states that the net magnetic flux through a closed surface is zero. Applying this to a thin wafer across the boundary (see the above figure),

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

which leads to the continuity condition:

$$B_{\perp,1} = B_{\perp,2}.$$

However, the perpendicular component of $\vec{H}$ is not necessarily continuous. From the relation $\vec{H} = \frac{1}{\mu_0} \vec{B} - \vec{M}$, we derive the boundary condition:

$$H_{\perp,1} - H_{\perp,2} = M_{\perp,2} - M_{\perp,1}.$$

Since $M_{\perp,2}$ and $M_{\perp,1}$ are not necessarily equal, $H_{\perp}$ may exhibit a discontinuity. For the previously discussed magnetized material, consider its two end faces. At these two boundaries, $M_{\perp}$ is nonzero inside the material but drops to zero immediately outside. This abrupt change results in a discontinuity in the perpendicular component of both $\vec{M}$ and $\vec{H}$.

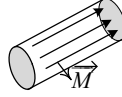


Figure. $M_{\perp}$ is discontinuous at the two end faces (shaded) of a magnetized rod.

Suppose we perform the same volume integral of $\vec{M}$ over the top or bottom end face of the magnetized rod, as shown below.

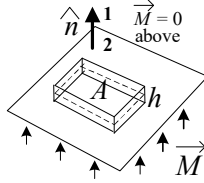


Figure. Schematic illustration for deriving boundary condition when $\vec{M}$ is discontinuous.

In this special case, the integral is no longer zero, $\oint_s \vec{M} \cdot d\vec{a} \neq 0$, which leads to $\oint_s \vec{H} \cdot d\vec{a} \neq 0$, and consequently $\nabla \cdot \vec{H} \neq 0$. In other words, while $\vec{B}$ remains divergence-free due to the absence of magnetic monopoles, $\vec{H}$ is not necessarily a divergence-free quantity.

2) Tangential (parallel) components

The magnetic field component parallel to the boundary is a vector that can be further decomposed into two components: one perpendicular to the surface current and the other parallel to it. For the perpendicular component, we evaluate the loop integral as illustrated in the left panel of the following figure.

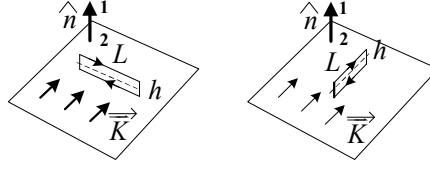


Figure. Schematic illustration for deriving boundary conditions for the tangential component. Left: Direction perpendicular to the current. Right: Direction parallel to the current.

The integral can be evaluated as $\oint_p \vec{B} \cdot d\vec{l} = (B_1 - B_2)L = \mu_0 K L$, which simplifies to $B_1 - B_2 = \mu_0 K$. For the parallel component, performing a loop integral as shown in the right panel of the figure yields $B_1 - B_2 = 0$, since the loop encloses no net current. These results can be unified into a single boundary condition for the tangential component of the magnetic field:

$$\vec{B}_{\parallel,1} - \vec{B}_{\parallel,2} = \mu_0 (\vec{K} \times \hat{n}),$$

where $\hat{n}$ is the unit normal pointing from region 2 to region 1. Similarly, the boundary condition for the parallel component of $\vec{H}$ is:

$$\vec{H}_{\parallel,1} - \vec{H}_{\parallel,2} = \vec{K}_f \times \hat{n}.$$

6. Nonlinear magnetic materials

Unlike linear magnetic materials, where the relationship between $\vec{H}$ and $\vec{B}$ is simple and proportional, nonlinear magnetic materials exhibit a more complex behavior where μ depends on H . This nonlinearity is particularly significant in ferromagnetic materials, where the magnetization response is highly dependent on the applied field. To explain the crazy behavior of nonlinear magnetic materials, quantum mechanics is needed. Here, we simply discuss some properties using ferromagnets as an example. Typical ferromagnetic materials include iron, cobalt, and nickel.

Suppose we magnetize an iron rod by placing it inside a current-carrying solenoid, as illustrated in the following figure.

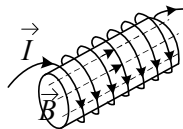


Figure. Schematic for magnetizing an iron rod.

We control the current I and geometric factors, such as the winding density n (the number of turns

per unit length of the solenoid). Using a method that we will discuss in the next section, we determine the H field inside the solenoid as $nI\hat{z}$, where $\hat{z}$ is the unit vector along the solenoid's axis. The corresponding magnetic field $\vec{B}$ inside the solenoid can be computed as:

$$\vec{B} = \mu_0(\vec{H} + \vec{M}),$$

where $\vec{M}$ is the induced magnetization. Here, $\vec{M}$ exhibits a much more complex dependence on $\vec{H}$ than the linear relationship we previously studied. When plotting the relationship between $\vec{B}$ and $\vec{H}$, we expect to observe a curve similar to the one shown in the following figure.

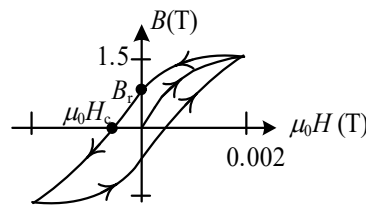


Figure. A typical hysteresis curve.

From this figure, we observe that the B - H curve is highly nonlinear, and forms a hysteresis loop, meaning that the material "remembers" past magnetization states even after the external field is removed. Remanence (B_r) is the residual magnetization when $H = 0$, while coercivity (H_c) is the required reverse field to demagnetize the material.

We also realize the fact that ferromagnetic materials exhibit very high magnetic susceptibility χ_m , typically in the range of 10^3 to 10^6 . This means that even a small applied field H can induce a large magnetization M (i.e., the x -axis reads 0.02 T while the y -axis reads 1.5 T), significantly amplifying the magnetic field B . This property is essential for electromagnets, transformers, and magnetic storage devices.

Ferromagnetic materials exhibit strong magnetization only below the Curie temperature T_c (770°C for iron). Above it, thermal agitation disrupts the alignment of atomic magnetic moments, and the material transitions to a paramagnetic state where magnetization disappears.

A microscopic explanation to the unusual behavior of ferromagnetic materials under an externally applied magnetic field lies in the presence of magnetic domains—small regions where atomic spins are aligned. When an external field is applied, these domains expand or rotate, leading to magnetization. The movement of domain walls plays a crucial role in the material's nonlinear response and the hysteresis effect.